CS-1101
PROGRAMMING FUNDAMENTALS

Lab 10

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Lab Contents

- Control Structures
- Repetition Control Structures
  - for Loop

Lab Objectives

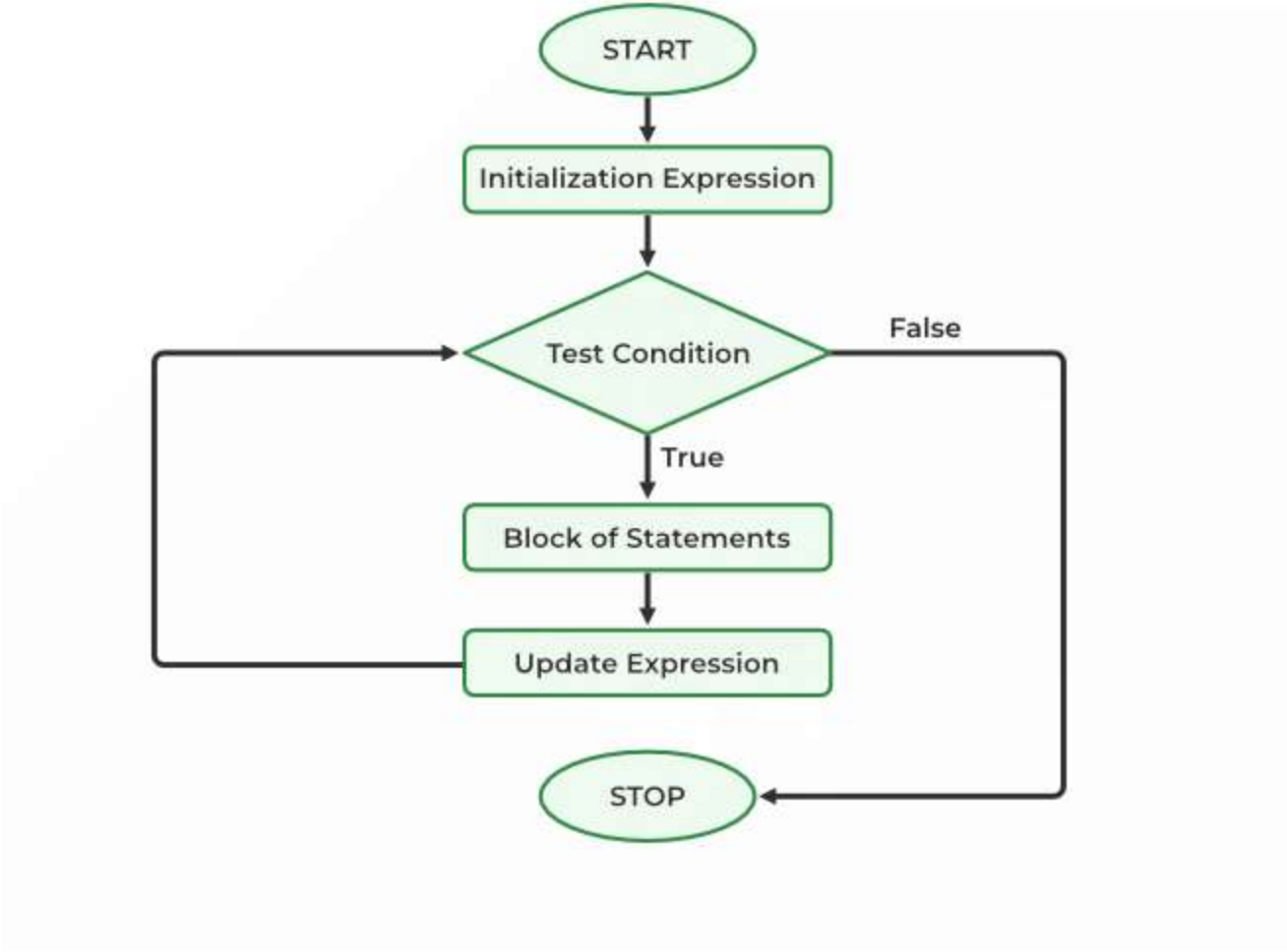
- The objective of this lab is to learn how to use for loop in C++ to perform repetitive tasks efficiently.

for Loop

- for loop is an **entry-controlled** loop that is used to execute a block of code repeatedly for the given number of times.
- It is generally preferred over while and do-while loops in case the number of iterations is known beforehand.

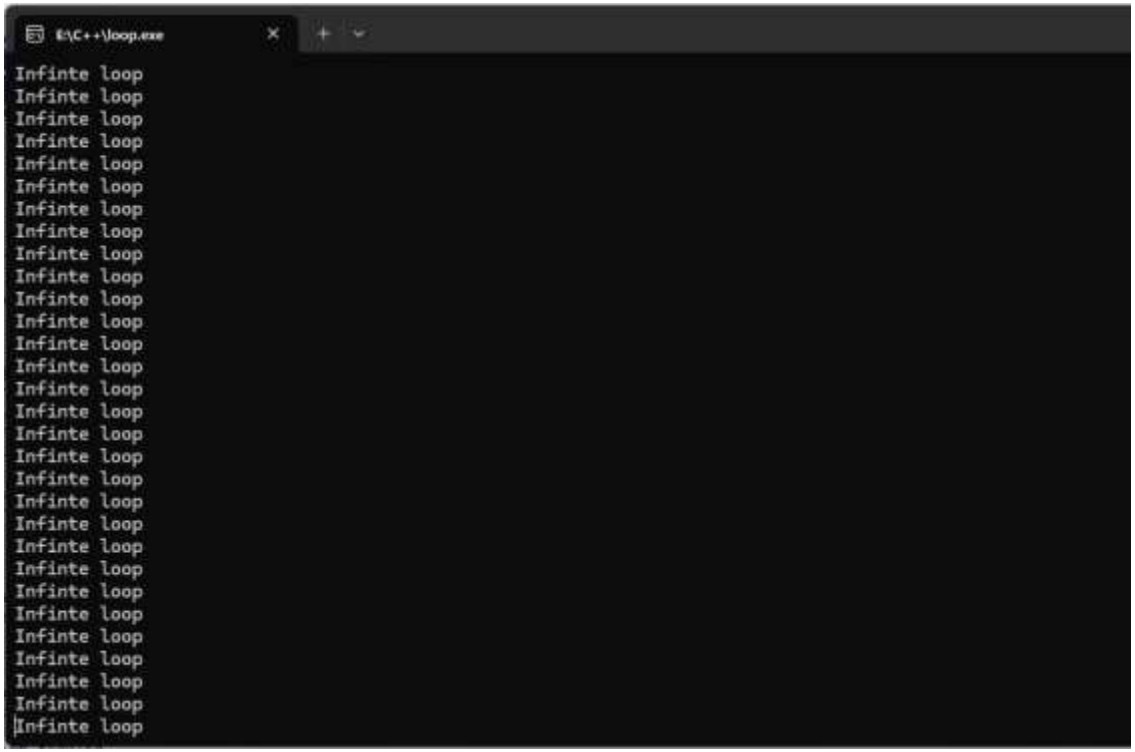
```
for ( initialization; test condition; updation) {
    // body of for loop
}
```

for Loop


START
Initialization Expression
Test Condition
False
True
Block of Statements
Update Expression
STOP

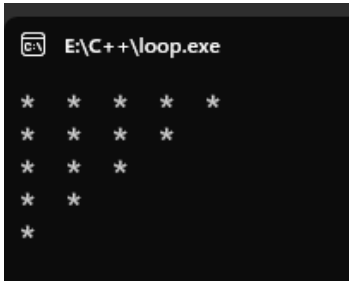
Loop that will run forever

```
for(;;){
    cout << "Infinte loop" << endl;
}
```



Tasks of for loop

- Write a program to print all numbers from 1 to N, where N is entered by the user.
- Write a program to display the multiplication table of a number entered by the user.
- Write a C++ program to calculate the sum of the first N natural numbers (1, 2, 3, 4, 5…N) where N is entered by the user.
- Write two programs that prints the following patterns.



- Write a program that asks the user to enter a number and then calculates the factorial of the entered number. If the user enters a negative number, display a message stating that the factorial of negative numbers is not defined.

References

- https://www.geeksforgeeks.org/cpp-for-loop/